

Long Hei Mak

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EDUCATION

University of Toronto

Toronto, ON

Honours Bachelor of Science in Computer Science and Economics – cGPA: 3.53/4.0

2019 – 2024

Relevant Coursework

- Algorithm Design and Analysis, Software Tools and Systems Programming, Data Structures and Analysis
- Computer Organization, Introduction to Software Engineering, Introduction to Databases

EXPERIENCE

Contract Software Developer - Technologies: Flutter, AWS, Go

Oct 2025 – Feb 2026

LuluHK Technologies Ltd.

Kowloon, Hong Kong

- Developed cross-platform mobile applications for iOS and Android using Flutter, delivering a seamless user experience across both platforms
- Designed and implemented core map-based features, including interactive map view, favourites logic, custom map markers, and map popups to significantly improve usability, visibility, and user engagement
- Spearheaded the end-to-end development of the Promotions feature, building reusable frontend components in Flutter and designing new data structures, models, and RESTful API endpoints in the backend to support future marketing campaigns
- Led production deployment and infrastructure setup on AWS, utilizing EC2 instances for application servers, S3 for static assets (images, files, and website hosting), and Amazon DocumentDB for scalable persistent data storage.

Teaching Assistant - Data Structures and Analysis

Jan 2022 – Apr 2022

University of Toronto Mississauga

Mississauga, ON

- Led weekly tutorial sessions of 15-20 students and guided student groups to complete the weekly problem set
- Graded assignments and exams and assessed students on their understanding of the course material

PROJECTS

Transit System Command-Line Simulator (Technologies: Java, SQL and Scrum Methodology)

- Developed a comprehensive command-line simulation of a public transit fare system in Java, enabling user registration, card balance management, tap-in/tap-out functionality at stations, and dynamic fare calculation based on travel routes and zones
- Implemented persistent data storage using SQLite (via JDBC), enabling saving and loading of user accounts, card balances, trip histories, station/line configurations, and system state between program sessions.
- Implemented core fare enforcement logic, including balance checks to prevent entry on insufficient funds, trip cost deduction upon exit, and support for transfers with time-based rules
- Designed and built extensible system administration features, allowing admins to add/modify transit lines, stations, and related infrastructure through command-line interfaces
- Collaborated in a 4 person team using Scrum methodology, participating in sprint planning, daily stand-ups, backlog grooming, sprint reviews, and retrospectives to deliver iterative project milestones on schedule

Smart Grocery App (Technologies: Flutter, Django, SQLite),

- Collaborated in a team of seven using Agile methodology to create a full-stack app to simplify grocery shopping for both companies and customers
- In charge of the Django backend for routing through a store and user registration and authentication and the Flutter frontend for mapping and displaying store inventory to the customer

Schmart App (Technologies: React Native, Flask) Hack the Valley - 1st Place (Devpost)

October 2021

- Collaborated in a team of four to create a smart shopping app using React Native. We designed and implemented a barcode scanner which scanned objects and queried a nutrition API to obtain the nutritional information about the object and dynamically updated in the app
- Worked on the user registration and login authentication in the frontend using React Native and backend using Flask, creating a system to register and log in

PROGRAMMING SKILLS

Languages: Dart, Python, Java, C, JavaScript, HTML/CSS

Frameworks: Flutter, React Native

Developer Tools: Git, Jira, Docker, VS Code, PyCharm